

Introduction to SIEM For SOC Analysts

Objective: To learn what a SIEM is, how it works, the uses, and why it is important to SOC Analysts.

1. What is a SIEM?

A SIEM (Security Information and Event Management) system is a security platform that collects, centralizes, analyzes, and correlates logs from different security systems or tools across an organization to detect security threats. Instead of checking logs manually on each device, a SOC analyst uses SIEM to monitor all the logs in a centralized dashboard.

2. Why SOC Analyst Depends on SIEM?

Modern IT infrastructure generates millions of logs per day from endpoints, servers, firewalls, and network devices. Without SIEM, detecting attacks would be slow, manual and inefficient.

SIEM helps SOC analyst to :

- Detect threat early
- Correlate activity across multiple systems
- Alerting
- Reduce response time to incidents.
- Quickly investigate alerts.

SIEM is the core tool used by SOC Analysts.

3. What Does a SIEM Do?

- Logs collection/aggregation: collect and gather logs from multiple sources across an organization.
- Logs Normalization: it converts different log formats into a common structure.
- Logs Correlation: Connect related events across different systems to identify suspicious patterns.
- Dashboarding: It manages and monitors the logs from various sources from a centralized dashboard.
- Alerting: Generates alert based on predefined rules or detected anomalies

4. SIEM Log Ingestion Flow

- **Logs Ingestion:** Logs are collected from endpoints, servers, firewalls, network devices, applications and cloud services. Logs are sent to SIEM in real time or near real time.

- **Correlation:** Connects multiple events from different sources to detect suspicious patterns. SIEM links multiple events together.
E.g 5 failed login attempts > Successful login > privilege escalation.
- **Alerting:** It generates alerts based on predefined rules or detection of anomalies. When correlation rules are triggered, an alert is generated. Alerts are assigned severity levels (Low, Medium, High).

5. Types of Logs Collected by SIEM.

SIEM tools collect many log types, including:

- Application Logs
- Network traffic logs: IDS/IPS alerts
- Firewall logs: Allowed or Blocked Connection
- Cloud logs: identity access, API activity
- Endpoints logs: Malware alerts, process activities, authentication logs
- Servers logs: Linux Syslog, Windows event viewer

This log helps the SOC analysts have a holistic view of an attack.

6. Why SIEM Reduces Investigation Time

- Logs are centralized in one place
- Correlated events reveal attack timelines
- Alerts point analysts directly to suspicious activity
- Investigations take minutes instead of hours

This significantly improves incident response speed and accuracy.

7. Common SIEM Tools Used in SOC

- Splunk: Widely used, powerful search and analysis
- Microsoft Sentinel : Cloud-native SIEM integrated with Azure
- IBM QRadar – Strong correlation and threat detection capabilities

SOC Analysts must understand the concept of SIEM even if tools differ.

Summary

A SIEM is the central nervous system of a SOC. It enables analysts to collect logs, correlate events, and detect threats efficiently. Mastering SIEM fundamentals is essential for anyone preparing for a SOC Analyst role.

This study improved my understanding of SIEM functionality, log correlation, and how SOC analysts use SIEM tools to reduce investigation and response.